

FretGarden

Pedagogical Constitution

Chapter 15

Ear-Training and Auditory Pedagogy

Version	1.0
Status	Foundational Draft
Document Type	Normative Pedagogical Standard
Authority	Binding on all FretGarden content, systems, and decisions within the scope of Chapter 15
Primary Dependencies	Foundation; Chapters 2 through 14

This chapter defines how FretGarden develops hearing as an active, testable, and musically connected capability.

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1. Purpose

This chapter governs auditory perception, discrimination, identification, audiation, imitation, singing, transcription, error detection, and the connection between hearing and fretboard action.

Ear training in FretGarden is not a detached quiz category. It is a system for making musical relationships perceptually available before, during, and after performance.

2. Constitutional Authority

This chapter specializes prior standards for auditory objectives and applies to audio examples, ear-training games, playback systems, response modes, assessment, review, and automated feedback.

3. Governing Principle

Governing Rule: Auditory learning MUST require listening to the target relationship and must progress toward recognition, reproduction, prediction, and use under varied musical conditions.

4. Core Distinctions

Hearing sound is not attending to it. Discrimination is not identification. Identification is not reproduction. Reproduction is not audiation. Familiarity is not transfer. Perfect pitch is not general musicianship.

5. Auditory Domain Model

The domain includes pitch direction, contour, interval, scale degree, chord quality, chord tone, cadence, bass motion, rhythm, meter, timbre, articulation, tuning, transcription, audiation, and performance monitoring.

6. Listening Conditions

Audio level, device quality, latency, noise, instrument timbre, register, and playback repetition affect evidence. Important assessments must record or control relevant conditions.

7. Attention Before Labeling

Learners should first know what feature to attend to: direction, distance, stability, chord quality, pulse, or resolution. Asking for a label without establishing the perceptual target encourages guessing.

8. Same-Different Discrimination

Early tasks may compare examples and isolate one dimension. These tasks provide limited evidence and should progress toward independent identification and reproduction.

9. Pitch Direction and Contour

Direction and contour form foundations for melodic hearing. Learners should track motion across varied registers and timbres rather than memorize one sound set.

10. Interval Hearing

Interval pedagogy should vary melodic versus harmonic presentation, direction, register, timbre, and tonal context. Familiar-song cues may support entry but must not remain the sole strategy.

11. Scale-Degree Hearing

Degrees should be heard relative to an established tonic. Tasks must verify that the tonic is perceived rather than assumed from an interface label.

12. Chord and Chord-Tone Hearing

Learners should distinguish quality, inversion, bass, function, and individual chord tones according to objective scope. Dense voicings and unfamiliar timbres should be introduced progressively.

13. Rhythm and Meter Hearing

Auditory rhythm tasks should require pulse extraction, imitation, continuation, transcription, and detection of displacement or duration changes.

14. Audiation

Audiation is internally hearing music without current external sound. Tasks may use silent continuation, prediction, mental transposition, and imagined resolution.

Self-report alone cannot establish audiation; a later produced or selected response is needed.

15. Singing and Vocalization

Singing can externalize hearing, but vocal accuracy is influenced by range, technique, confidence, and accessibility. Feedback must separate perceptual understanding from vocal execution when appropriate.

16. Imitation

Imitation connects perception to production. Examples should be short enough for attention, then lengthen and vary as memory and control improve.

17. Transcription

Transcription integrates hearing, memory, representation, and reasoning. Errors must be classified by dimension rather than treated as a single failure.

18. Instrument-to-Ear Mapping

Learners should hear a target, find it on the instrument, compare, and adjust. Visual cues should be withheld when auditory search is the objective.

19. Ear-to-Theory Mapping

Labels should explain perceived relationships. Theory tasks that can be solved without listening do not count as ear training even when audio is present.

20. Context and Tonal Center

Auditory identification depends on context. The system should vary isolated, tonal, modal, harmonic, and musical examples while clearly specifying the claim each supports.

21. Timbre and Register Variation

Transfer requires examples beyond one synthesized piano timbre and one register. Variation should be staged so that timbre change does not overwhelm the target relation.

22. Repetition and Replay

Replay may support learning but changes evidence conditions. Assessments must limit or record replay count. Unlimited replay cannot support the same claim as one-hearing recognition.

23. Guessing Controls

Selected-response tasks should use plausible distractors, confidence checks, constructed follow-ups, and varied item order. Correct chance responses must not be treated as stable knowledge.

24. Feedback Timing

Immediate feedback can recalibrate perception, but complete answer revelation should be followed by comparison and later unsupported retrieval.

25. Musical Application

Auditory skills should guide tuning, phrase imitation, chord-tone targeting, transcription, improvisation, error detection, and repertoire learning.

26. Practice Architecture

Ear training should use short, attentive sets with varied tasks and stopping rules for fatigue. Repeated inattentive guessing degrades evidence and motivation.

27. Assessment

Assessment should define number of hearings, response mode, context, timbre, register, pitch reference, and chance level. Strong claims require delayed and varied evidence.

28. Mastery

Auditory mastery is scoped by relation type, context, support, timbre, register, response mode, retention, and transfer. A global 'good ear' label is noncompliant.

29. Review

Review should vary examples while preserving target structure. Old recordings should not dominate to the point that learners recognize the recording rather than the relation.

30. Error Classification

Errors include attention failure, direction confusion, interval-class confusion, tonic instability, chord-quality confusion, inversion error, rhythm-memory loss, timbre interference, response-mapping error, and random guessing.

31. Remediation

Remediation may contrast confusable examples, re-establish tonic, reduce register span, change timbre, add singing, slow playback, isolate bass, or return to same-different discrimination.

32. Adaptive Difficulty

Variables include item set, context, register, timbre, presentation type, number of hearings, response format, sequence length, harmonic density, and time between sound and response.

33. Technology and Latency

Playback and recording systems must account for latency, compression, device routing, and microphone quality where timing or intonation is assessed. System faults must not become learner evidence.

34. Accessibility

Equivalent modes may use vibration, visual spectrographic support, adaptive hearing settings, transposition, or non-vocal responses when they preserve the target. Some auditory objectives may not have full non-auditory equivalents; claim limits must remain honest.

35. Emotional Safety

Singing and ear tests can create vulnerability. Feedback must avoid shame, talent myths, and fixed-ability language. Difficulty should be framed as trainable perceptual calibration.

36. Anti-Patterns

Noncompliant patterns include song mnemonics as final mastery, one timbre only, unlimited replay without recording support, visual answers during auditory tasks, global ear scores, treating vocal mismatch as perceptual failure, and rapid guessing loops.

37. Constitutional Non-Negotiables

1. Ear-training tasks MUST make listening necessary.
2. Hearing count and support conditions MUST be recorded for evidence claims.
3. Recognition MUST progress toward reproduction or application where appropriate.
4. Timbral and contextual variation MUST support broad claims.
5. Guessing MUST be controlled.
6. Vocal execution MUST not be conflated automatically with hearing.
7. Technology faults MUST not count as negative learner evidence.
8. Auditory work MUST connect to musical action.

38. Conformance Tests

Reviewers must identify the audible target, listening conditions, attention cue, response mode, chance level, replay policy, feedback loop, transfer task, and review plan.

39. Tradeoffs and Resolution Rules

Isolation versus context: isolate for diagnosis, then restore musical context.

Replay support versus evidence strength: permit learning support but report the condition.

Vocalization versus accessibility: use singing when useful and provide valid alternatives where the target allows.

40. Assumptions Made

This chapter assumes most learners do not possess reliable absolute pitch and that relative hearing is the principal instructional pathway.

41. Unresolved Questions

Unresolved matters include automated pitch and rhythm tolerances, headphone calibration, hearing-profile settings, default replay limits, and validity of confidence-weighted scoring.

42. Topics Intentionally Deferred

Clinical hearing assessment, advanced audio engineering, absolute-pitch training, and specialist dictation curricula are deferred.

43. Required Future Cross-References

Harmony, improvisation, composition, repertoire, analytics, accessibility, and automated-feedback specifications must preserve these evidence conditions.

44. Chapter Conformance Statement

A compliant auditory system develops attentive, discriminating, reproducible, retained, and musically actionable hearing while keeping claims proportional to actual listening conditions.